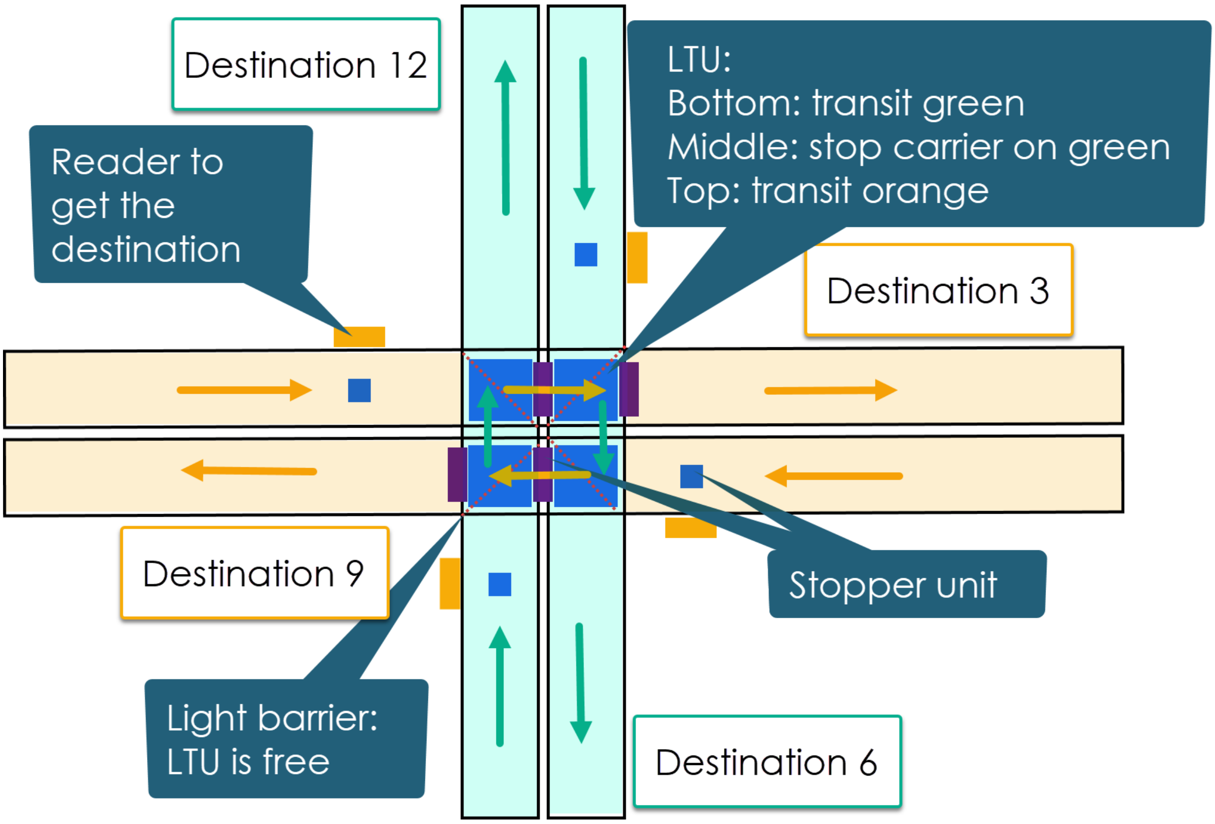
ADVANCED BECKHOFF TWINCAT TASK

Brief Description

There is conveyor interchange to send carrier to the next process step.



Conveyor

The Conveyors are like TS2+ conveyors, each of the four conveyors has just one motor with one boolean output to turn it on and two boolean inputs, conveyor is running, and motor is OK (fuse, thermistor).

Conveyor tracks:

1. From 12 to 6: lower track (green)
2. From 6 to 12: lower track (green)
3. From 3 to 9: upper track (orange)
4. From 9 to 3: upper track (orange)

LTU

The LTU (Lift-Transfer-Unit) has a cylinder with two outputs and three inputs:

Output to Bottom	Output to Top	Input at Bottom	Input at Middle	Input at Top
FALSE	FALSE	FALSE	TRUE	FALSE
FALSE	TRUE	FALSE	FALSE	TRUE
TRUE	FALSE	TRUE	FLASE	FALSE

Each LTU has one light barrier to check if it's free. The light barriers are mounted from the inner corner to the outer corner.

For the transfer at Top, each LTU has one motor. The motor interface is equal to the conveyor motor.

Stopper unit

Each stopper unit has one cylinder with one output (TRUE is move stopper down) and zero inputs.

Each stopper unit (blue and purple) has its own sensor to check if the stopper is occupied.

Reader

The reader could be any reader, use here just a dummy function which returns randomly the numbers 12, 3, 6 or 9.

Your Task

Implement an automatic sequence to control the carrier routing on the interchange with the focus on throughput and that every carrier will be processed. If you still have time and desire, you can still implement the homing sequence.

It must be solved in **ST** or **SFC**. Please use **TwinCAT Version 2 or 3.**

If any instruction is unclear or under-specified, make an assumption about it and document it in your code.